

Deep Learning-Based Classification of Imbalanced Wireless Capsule Endoscopy (WCE) Dataset

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Abstract

This project focuses on the classification of gastrointestinal diseases using deep learning models on the imbalanced Kvasir-Capsule dataset. Various techniques such as under-sampling, data augmentation, transfer learning, fine-tuning, and adaptive learning rate control were applied to improve classification performance. The study evaluates multiple architectures and compares their effectiveness under different training conditions.

1 Introduction

Medical image classification plays a crucial role in early disease detection. However, datasets in this domain often suffer from class imbalance, which can lead to biased models. This project addresses these challenges using deep learning techniques on the Wireless Capsule Endoscopy dataset.

2 Dataset Description

The Kvasir-Capsule dataset consists of gastrointestinal tract images categorized into multiple classes. The dataset exhibits significant imbalance, where some classes contain many samples while others have very few.

3 Methodology

3.1 Data Preprocessing

- Images resized to 224×224
- Pixel values normalized to $[0,1]$
- Dataset split into:
 - Training: 70%

- Validation: 15%
- Testing: 15%

3.2 Handling Class Imbalance

- Random under-sampling applied to majority classes
- Data augmentation applied to minority classes:
 - Horizontal flip
 - Rotation
 - Zoom and cropping

3.3 Model Design

The following pre-trained models were used:

- ResNet18
- MobileNetV2
- EfficientNet-B0

3.4 Transfer Learning

- Pre-trained ImageNet weights used
- Early layers frozen
- Final layers replaced for classification

3.5 Fine-Tuning

- ResNet18 was fine-tuned by unfreezing deeper layers
- Improved adaptation to medical images

3.6 Learning Rate Strategies

- Reduce on Plateau
- Cosine Annealing
- Constant Learning Rate

3.7 Loss Function

- Class-weighted cross-entropy loss used
- Helps prioritize minority classes

4 Experimental Setup

- Optimizer: Adam
- Epochs: 10
- Batch size: 32
- Evaluation metrics:
 - Accuracy
 - Precision
 - Recall
 - F1-score

4.1 Observations

- Fine-tuned ResNet performed best due to better feature adaptation
- EfficientNet showed strong performance but required more training
- MobileNet was faster but slightly less accurate

5 Confusion Matrix Analysis

Confusion matrices were used to analyze class-wise performance. The model showed improved classification of minority classes after applying augmentation and weighted loss.

6 Discussion

- Class imbalance significantly affects model performance
- Combining data-level and algorithm-level methods improves results
- Learning rate strategies impact convergence behavior

7 Novel Contributions

- Class-weighted loss for imbalance handling
- Partial fine-tuning of deep layers
- Comparison of multiple learning rate strategies
- Comprehensive evaluation using multiple metrics

8 Conclusion

The project demonstrates that combining under-sampling, augmentation, transfer learning, and adaptive learning strategies significantly improves classification performance on imbalanced medical datasets.

9 Future Work

- Use larger datasets for better generalization
- Explore EfficientNet-V2 and advanced architectures
- Apply attention mechanisms
- Use Grad-CAM for model interpretability